

THE RECIPROCAL SUM OF THE PRIME-PREFIX-FREE INTEGERS CONVERGES UNDER THE RIEMANN HYPOTHESIS

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ABSTRACT. Call a positive integer *prime-prefix-free* if none of its proper binary prefixes $\lfloor n/2^k \rfloor$, $k \geq 1$, is an odd prime (OEIS A287117). Assuming the Riemann Hypothesis, we prove that the sum of $1/n$ over the prime-prefix-free integers converges. More precisely, the proportion of $[2^m, 2^{m+1})$ they occupy is $O_\delta(m^{-1/\log 2 + \delta})$ for every $\delta > 0$. The heuristic exponent is $1/\log 2 \approx 1.4427$, and convergence rests on the inequality $1/\log 2 > 1$, that is, on the base 2 being smaller than e . The proof combines Selberg's theorem on primes in almost all short intervals with an upper-bound sieve for prime pairs, and closes with a bootstrap for a delayed recurrence. The convergence statement is machine-verified in Lean 4 over Mathlib from Mathlib's statement of the Riemann Hypothesis. Numerically the sum exceeds 3.5374 unconditionally, and a heuristic model fitted to exact counts up to 2^{46} puts it near 3.877.

1. INTRODUCTION

Arrange the positive integers in the infinite binary tree in which the children of n are $2n$ and $2n + 1$. The ancestors of n are then its proper binary prefixes $\lfloor n/2^k \rfloor$, $k \geq 1$. Let S be the set of n none of whose ancestors is an odd prime. Equivalently, the odd primes are made leaves of the tree, and S is the set of nodes that remain. The prime 2 is excluded because every $n \geq 4$ has 2 or 3 as a prefix, so including it would leave a finite set. The sequence $S = \{1, 2, 3, 4, 5, 8, 9, 16, 17, 18, 19, 32, \dots\}$ is A287117 in the OEIS [7], and the question whether $\sum_{n \in S} 1/n$ converges was posed on Mathematics Stack Exchange [6].

Theorem. *Assume the Riemann Hypothesis. Then $\sum_{n \in S} 1/n < \infty$. Moreover, for every $\delta > 0$,*

$$\#(S \cap [2^m, 2^{m+1})) \ll_\delta 2^m m^{-1/\log 2 + \delta}.$$

The exponent is the one predicted by a probabilistic model. Pick $x \in [1, 2)$ at random. Its binary prefix of length $m + 1$ is an integer near 2^m , prime with probability about $1/(m \log 2)$, and these events are nearly independent, so the prefixes of x avoid the odd primes up to length m with probability about $\prod_{k \leq m} (1 - 1/(k \log 2)) \asymp m^{-1/\log 2}$. That survival probability is the proportion r_m of the level $[2^m, 2^{m+1})$ occupied by S , and the reciprocal sum is comparable to $\sum_m r_m$. Convergence therefore hinges on $1/\log 2 \approx 1.4427$ exceeding 1. In base b the same model gives the exponent $1/\log b$, below 1 for every $b \geq 3$: binary is the one base in which the analogous sum is expected to converge.

Turning the model into a proof requires lower bounds for primes in the set of survivors at every level. That set is a union of short intervals at all scales, so this is a problem about primes in short intervals, counted simultaneously along all branches of the tree. The proof uses two known results. The first is Selberg's theorem [8], conditional on RH, that almost all intervals $[x, x + h]$ with h a small power of x contain the expected number of primes; the exceptional set is small enough to survive a union over the polynomially many scales involved. The second is the classical upper-bound sieve for prime pairs [4]. It controls primes that have a prime ancestor only a few levels up. These pairs are beyond the reach of RH, and the sieve overcounts them by a constant factor, but they occupy only a small fraction of the levels, and that fraction can be made as small as we like. A telescoping identity converts the remaining long-range

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pairs into the same quantities r_m that the argument controls, and a bootstrap for the resulting delayed recurrence finishes the proof.

Each loss in the argument lowers the exponent $1/\log 2$, and the proof needs the exponent to stay above 1, so the losses must total less than $1/\log 2 - 1 \approx 0.44$. In particular, the sieve's constant-factor overcount can be tolerated only on a small proportion of levels.

Notation. Logarithms are natural. For $m \geq 0$ let $I_m = [2^m, 2^{m+1}) \cap \mathbb{Z}$, $\mathcal{S}_m = \mathcal{S} \cap I_m$, P_m the set of odd primes in $\mathcal{S}_m$, and

$$r_m = |\mathcal{S}_m|/2^m \in [0, 1].$$

For integers $M \geq 1$ and $b \geq 0$ let $B(M, b) = [M2^b, (M+1)2^b) \cap \mathbb{Z}$. These are the descendants of M lying b levels below it: $n \in B(M, b)$ if and only if $\lfloor n/2^b \rfloor = M$. For a finite set A of integers write $\pi(A)$ for the number of primes in A and $\vartheta(A) = \sum_{p \in A} \log p$. As usual $\vartheta(x) = \sum_{p \leq x} \log p$. Throughout, $c = 1/\log 2$.

2. COMPONENTS

Lemma 2.1 (Selberg [8]). *Assume the Riemann Hypothesis. For every $\theta > 0$ there is C such that for all integers $X \geq 2$ and h with $X^\theta \leq h \leq X^{1/2}$,*

$$\sum_{X \leq n < 2X} (\vartheta(n+h) - \vartheta(n) - h)^2 \leq C h X \log^2 X.$$

Selberg proved the integral form $\int_X^{2X} (\psi(x+h) - \psi(x) - h)^2 dx \ll h X \log^2 X$ for $1 \leq h \leq X$. For integer h the window $\vartheta(x+h) - \vartheta(x)$ is constant on each $[n, n+1)$, so the integral of the ϑ -version is the sum above, and the passage from ψ to ϑ costs only the prime powers p^k , $k \geq 2$, which contribute $O(h X \log^2 X)$ to the mean square. Lemma 2.1 asks for much less than Selberg's theorem. Only windows of length at least X^θ are needed, for a θ we will take small but fixed.

Lemma 2.2 (Upper-bound sieve for prime pairs; see [4]). *There is an absolute constant C such that for all integers $Y \geq 0$, $H \geq 2$, $a \geq 1$, $t \geq 1$,*

$$\#\{Y \leq n < Y+H : n \text{ and } n+t \text{ prime}\} \leq C \left(\frac{at}{\varphi(at)} \right)^2 \frac{H}{\log^2 H}.$$

The bound is uniform in a and t . The sieve only sifts by primes not dividing $2at$, and the factor $(at/\varphi(at))^2$ absorbs the corresponding local densities.

Lemma 2.3. $\sum_{t \leq T} (t/\varphi(t))^2 \ll T$.

Proof. Write $(t/\varphi(t))^2 = \sum_{d|t} g(d)$ with g multiplicative, supported on squarefree d , and $g(p) = (p/(p-1))^2 - 1 \leq 6/p$. Then the sum is $\sum_d g(d) \lfloor T/d \rfloor \leq T \prod_p (1 + 6/p^2)$. $\square$

3. THE TREE

Lemma 3.1. (i) *For $e \in \{0, 1\}$, $2n+e \in \mathcal{S}$ if and only if $n \in \mathcal{S}$ and n is not an odd prime.*
(ii) *If $n \in \mathcal{S}$ then $\lfloor n/2^d \rfloor \in \mathcal{S}$ for every $d \geq 0$.*
(iii) $|\mathcal{S}_{m+1}| = 2(|\mathcal{S}_m| - |P_m|)$. Hence $r_{m+1} = r_m - |P_m|/2^m$, and r is nonincreasing.
(iv) $\sum_{n \in \mathcal{S}} 1/n \leq \sum_{m \geq 0} r_m$.

Proof. The proper prefixes of $2n+e$ are n and the proper prefixes of n , which gives (i); (ii) holds because the prefixes of $\lfloor n/2^d \rfloor$ are prefixes of n ; (iii) follows from (i); and (iv) from $1/n \leq 2^{-m}$ on I_m . $\square$

By (iii), $r_m - r_{m+1}$ is the number of odd primes in $\mathcal{S}_m$ divided by 2^m , so upper bounds for r come from lower bounds for these primes.

Lemma 3.2 (Counting inequality). *For integers $j \geq 2$ and $1 \leq d \leq j$,*

$$\sum_{N \in \mathcal{S}_{j-d}} \pi(B(N, d)) \leq |P_j| + \sum_{b=1}^d \sum_{p \in P_{j-b}} \pi(B(p, b)).$$

Proof. The blocks on the left are disjoint subsets of I_j , so the left side counts primes $q \in I_j$ with $\lfloor q/2^d \rfloor \in \mathcal{S}$. Such a q is at least $2^j \geq 4$, hence odd. If $q \in \mathcal{S}$, then $q \in P_j$. Otherwise some $\lfloor q/2^k \rfloor$ with $k \geq 1$ is an odd prime. For $k > d$ that prefix is a proper prefix of $\lfloor q/2^d \rfloor \in \mathcal{S}$, so $k \leq d$. Let $b \leq d$ be the largest such k and $p = \lfloor q/2^b \rfloor$. Every proper prefix of p is $\lfloor q/2^{b+k'} \rfloor$ with $k' \geq 1$. For $b + k' \leq d$ it is not an odd prime by the maximality of b ; for $b + k' > d$ it is a prefix of an element of $\mathcal{S}$. So $p \in P_{j-b}$ and $q \in B(p, b)$. $\square$

Lemma 3.2 bounds $|P_j|$ from below by the block counts on the left, less the sum over b , which counts the primes with an odd-prime ancestor at most d levels up. We split that sum at a threshold $b_0 = \lceil \theta j \rceil$. Long gaps $b \geq b_0$ are handled by Lemma 2.1 in Section 4, and short gaps $b < b_0$ by Lemma 2.2 in Section 5.

4. BLOCKS

Call a block $B(M, b)$ with $M \in I_{j-b}$ ε -bad if $|\vartheta(B(M, b)) - 2^b| > \varepsilon 2^b$.

Lemma 4.1 (Few bad blocks). *Assume Lemma 2.1. Let $\theta > 0$ and $0 < \varepsilon \leq 1$. There are C and j_1 such that for all $j \geq j_1$ and all b with $\theta j \leq b \leq j/2$, the number of ε -bad blocks $B(M, b)$, $M \in I_{j-b}$, is at most $C j^3 2^j / 4^b$.*

Proof. Let $X = 2^j$, $h = 2^b$ and $E(n) = \vartheta(n + h) - \vartheta(n) - h$. The hypotheses give $X^\theta \leq h \leq X^{1/2}$. For $M \in I_{j-b}$ we have $\vartheta(B(M, b)) - h = E(Mh - 1)$. Put $L = \lfloor \varepsilon h / (4 \log 3X) \rfloor$. Since $h \geq X^\theta$, for $j \geq j_1$ the quantity $\varepsilon h / (4 \log 3X)$ is at least 2, so $L \geq \varepsilon h / (8 \log 3X) \geq 1$. For $Mh \leq n < Mh + L$, the difference $E(n) - E(Mh - 1)$ is the ϑ -mass of two intervals of at most L integers below $3X$, so it is at most $2L \log 3X \leq \varepsilon h / 2$ in absolute value. Hence a bad block yields L consecutive integers $n \in [X, 2X]$ with $|E(n)| > \varepsilon h / 2$, and these runs are disjoint for distinct M . By Lemma 2.1,

$$\#\{\text{bad } M\} \cdot L \cdot \frac{\varepsilon^2 h^2}{4} \leq \sum_{X \leq n < 2X} E(n)^2 \leq C_1 h X \log^2 X,$$

so the number of bad blocks is at most $32C_1 X \log^2 X \log 3X / (\varepsilon^3 h^2) \ll j^3 2^j / 4^b$. $\square$

Lemma 4.2 (Good blocks). *Let $j \geq 1$, $b \leq j$ and $M \in I_{j-b}$. If $B(M, b)$ is not ε -bad, then*

$$\frac{(1 - \varepsilon) 2^b}{(j + 1) \log 2} \leq \pi(B(M, b)) \leq \frac{(1 + \varepsilon) 2^b}{j \log 2}.$$

In every case $\pi(B(M, b)) \leq 2^b$.

Proof. $B(M, b) \subset I_j$, so every prime p in it satisfies $j \log 2 \leq \log p < (j + 1) \log 2$. $\square$

5. SHORT GAPS

Lemma 5.1. *Assume Lemma 2.2. There is a constant C_5 such that for all integers $1 \leq b \leq d'$ with $b + d' \leq j$,*

$$\sum_{p \in P_{j-b}} \pi(B(p, b)) \leq C_5 |\mathcal{S}_{j-b-d'}| \frac{2^{b+d'}}{d'^2}.$$

Proof. The primes in $B(p, b)$ are $q = 2^b p + t$ with $0 \leq t < 2^b$, and t is odd since $q > 2$. By Lemma 3.1(ii), $N' = \lfloor p/2^{d'} \rfloor$ lies in $\mathcal{S}_{j-b-d'}$, and $p \in B(N', d')$, an interval of length $H = 2^{d'}$. Therefore the left side is at most

$$\sum_{N' \in \mathcal{S}_{j-b-d'}} \sum_{\substack{t < 2^b \\ t \text{ odd}}} \#\{n \in B(N', d') : n \text{ and } 2^b n + t \text{ prime}\}.$$

For odd t , $2^b t / \varphi(2^b t) = 2t / \varphi(t)$, so Lemma 2.2 with $a = 2^b$ bounds each count by

$$4C \left(\frac{t}{\varphi(t)} \right)^2 \frac{2^{d'}}{(d' \log 2)^2},$$

and Lemma 2.3 bounds the sum over t by a constant times 2^b . $\square$

Heuristically the left side is $|P_{j-b}| \cdot 2^b / (j \log 2)$, which in terms of r is of order $r 2^j / j^2$, the same order as the right side with $d' \asymp j$. The bound is therefore off by a constant factor, and for this reason the argument applies it only for $b < \theta j$, a small proportion of the levels.

6. THE RECURRENCE

Lemma 6.1. *Assume the Riemann Hypothesis. Let C_5 be as in Lemma 5.1 and let $0 < \varepsilon \leq 1$, $0 < \theta \leq 1/7$. There are K and j_0 such that for all $j \geq j_0$,*

$$r_{j+1} \leq r_j \left(1 - \frac{\varepsilon}{j} \right) + \left(\frac{\eta}{j} + \frac{K}{j^2} \right) r_{\lfloor j/3 \rfloor} + K j^3 2^{-\theta j}, \quad \eta = 2c\varepsilon + 5C_5\theta.$$

Proof. Let $d = \lfloor j/2 \rfloor$ and $b_0 = \lceil \theta j \rceil$. For j large, $1 \leq b_0 \leq j/6$, so $j - b - d \geq \lfloor j/3 \rfloor$ for every $b < b_0$. Apply Lemma 3.2 with this d and split the sum over b at b_0 .

Main term. Every $N \in \mathcal{S}_{j-d}$ whose block is not bad contributes at least $(1 - \varepsilon)2^d / ((j+1) \log 2)$ by Lemma 4.2. Lemma 4.1 with $b = d$ removes at most a proportion $O(j^3 2^{-d})$ of the 2^{j-d} blocks. So the left side of Lemma 3.2, divided by 2^j , is at least $(1 - \varepsilon) r_{j-d} / ((j+1) \log 2) - O(j^3 2^{-d})$.

Long gaps, $b_0 \leq b \leq d$. By Lemma 4.2 for good blocks, the trivial bound for bad ones, and Lemma 4.1,

$$\sum_{p \in P_{j-b}} \pi(B(p, b)) \leq |P_{j-b}| \frac{(1 + \varepsilon)2^b}{j \log 2} + O\left(\frac{j^3 2^j}{2^b}\right).$$

By Lemma 3.1(iii), $|P_{j-b}| 2^b = 2^j (r_{j-b} - r_{j-b+1})$, so the main parts telescope:

$$\frac{1}{2^j} \sum_{b=b_0}^d \sum_{p \in P_{j-b}} \pi(B(p, b)) \leq \frac{(1 + \varepsilon)(r_{j-d} - r_{j-b_0+1})}{j \log 2} + O(j^3 2^{-\theta j}).$$

Short gaps, $1 \leq b < b_0$. Lemma 5.1 with $d' = d$, together with $r_{j-b-d} \leq r_{\lfloor j/3 \rfloor}$, bounds their total, divided by 2^j , by $C_5 b_0 r_{\lfloor j/3 \rfloor} / d^2 \leq (5C_5 \theta / j + 5C_5 / j^2) r_{\lfloor j/3 \rfloor}$ for j large.

Assembly. Write $A = r_{j-d}$ and $B = r_{j-b_0+1} \geq r_j$. Lemma 3.2 and the three estimates give

$$\frac{|P_j|}{2^j} \geq \frac{(1 - \varepsilon) \frac{j}{j+1} A - (1 + \varepsilon)(A - B)}{j \log 2} - \left(\frac{5C_5 \theta}{j} + \frac{5C_5}{j^2} \right) r_{\lfloor j/3 \rfloor} - O(j^3 2^{-\theta j}).$$

The numerator equals $(1 + \varepsilon)B - (2\varepsilon + \frac{1-\varepsilon}{j+1})A \geq r_j - (2\varepsilon + 1/j) r_{\lfloor j/3 \rfloor}$, using $A \leq r_{\lfloor j/3 \rfloor}$. Inserting this into $r_{j+1} = r_j - |P_j|/2^j$ gives the claim. $\square$

In the numerator, the long-gap correction $(1 + \varepsilon)(A - B)$ nearly cancels the main term $(1 - \varepsilon)A$, leaving about $r_{j-b_0+1} \geq r_j$. Heuristically, the primes at level j below a surviving node d levels up number about $r_{j-d} 2^j / (j \log 2)$, and those removed by an earlier prime ancestor account for the decrease of r between levels $j - d$ and $j - b_0$. The cancellation holds up to the factors $1 \pm \varepsilon$, which is why the long gaps need the asymptotic count of Lemma 2.1 and cannot be handled by the sieve.

Lemma 6.2 (Bootstrap). *Let $r_j \geq 0$, $c, \eta \geq 0$, $\alpha \geq 0$ with $\alpha + 4^\alpha \eta \leq c$, and let $e_j, g_j \geq 0$ with $\sum e_j < \infty$ and $\sum j^\alpha g_j < \infty$. If*

$$r_{j+1} \leq r_j \left(1 - \frac{c}{j}\right) + \left(\frac{\eta}{j} + e_j\right) r_{\lfloor j/3 \rfloor} + g_j$$

for all large j , then $r_j = O(j^{-\alpha})$.

Proof. Let $s_j = j^\alpha r_j$ and $S_j = \max_{1 \leq i \leq j} s_i$. For $j \geq \max(12, c)$ we have $1 - c/j \geq 0$ and $\lfloor j/3 \rfloor \geq j/4$, so $r_{\lfloor j/3 \rfloor} \leq 4^\alpha S_j j^{-\alpha}$. Using $(1 + 1/j)^\alpha \leq e^{\alpha/j}$ and $1 + x \leq e^x$,

$$s_{j+1} \leq S_j e^{\alpha/j} \left(1 - \frac{c}{j} + \frac{4^\alpha \eta}{j} + 4^\alpha e_j\right) + (j+1)^\alpha g_j \leq S_j e^{4^\alpha e_j} + (j+1)^\alpha g_j.$$

Hence $S_{j+1} \leq S_j e^{4^\alpha e_j} + (j+1)^\alpha g_j$, and S_j is bounded. $\square$

Proof of the Theorem. Given $\delta > 0$, set $\alpha = c - \delta$ and choose ε and θ so small that $4^\alpha (2c\varepsilon + 5C_5\theta) \leq \delta$. Lemma 6.1 supplies the hypothesis of Lemma 6.2 with $e_j = K/j^2$ and $g_j = Kj^3 2^{-\theta j}$. So $|S_m| = 2^m r_m \ll 2^m m^{-c+\delta}$. Taking $\delta < c - 1$, the sum $\sum r_m$ converges, and Lemma 3.1(iv) finishes the proof. $\square$

For the convergence statement alone, the formalization fixes $\alpha = 6/5$, $\varepsilon = 1/200$ and $\theta = 1/(1000(C_5 + 1))$. Then $\eta \leq 0.02$, $4^{6/5} < 6$ and $6/5 + 6 \cdot 0.02 < 1.44 < c$.

7. REMARKS

Remark 7.1 (Where RH enters). *The Riemann Hypothesis is used once, through Lemma 2.1, and only for windows $h \geq X^\theta$ with θ small. The smallness of θ is forced by the sieve constant: the short-gap term contributes $5C_5\theta$ to η , and η has to fit inside the margin $c - 1 \approx 0.44$. Unconditionally, Lemma 2.1 is known for $\theta > 1/6$ by Huxley’s zero-density estimate [5], and for $\theta > 2/15$ after Guth and Maynard [3]. Both are far too large, since an upper-bound sieve cannot count the pairs $(p, 2p + 1)$, $(p, 4p + 1)$, $\dots$ within a factor close to 1 (the parity problem). A proof with no hypothesis would therefore need either primes in almost all intervals $[x, x + x^\theta]$ for a small explicit θ , or a way around the short gaps. The density hypothesis for ζ would serve in place of RH: it gives almost-all asymptotics for windows of length X^θ for every $\theta > 0$, with exceptional sets small enough for a weaker form of Lemma 4.1 that the argument tolerates.*

Remark 7.2 (Other bases). *In base b the same tree has branching b , a prefix at level m is prime with probability about $1/(m \log b)$, and the model gives $r_m \approx m^{-1/\log b}$. For $b \geq 3$ the exponent is below 1 and the analogous reciprocal sum is expected to diverge. The proof above uses the size of the exponent only through $c > 1$.*

Remark 7.3 (Numerics). *NUMERICS*

Remark 7.4 (Formal verification). *The convergence statement of the Theorem has been machine-verified in Lean 4 over Mathlib, through the bound $r_m \ll m^{-6/5}$, from Mathlib’s definition of the Riemann Hypothesis and with no other hypothesis beyond Lean’s three standard axioms. The statement, verbatim from PPF/Main.lean of the formalization (the file contains nothing else besides imports and a docstring), is:*

```
def PrimePrefixFree (n : ℕ) : Prop :=
  ∀ k : ℕ, 1 ≤ k → ¬ ((n / 2 ^ k).Prime ∧ n / 2 ^ k ≠ 2)

open Classical in
theorem ppf_summable_of_RH (hRH : RiemannHypothesis) :
  Summable (fun n : ℕ => if PrimePrefixFree n then (1 : ℝ) / n else 0) :=
  (PPF.summable_of_inputs (PPF.selbergMeanSquare_of_RH hRH) PPF.pairSieve).congr
  fun _ => if_congr Iff.rfl rfl rfl
```

Here `RiemannHypothesis`, `Prime` and `Summable` are `Mathlib`'s, and natural-number division $n / 2 \wedge k$ is the floor. The term $n = 0$ contributes $1/0 = 0$ by Lean's convention.

The proof has three audited layers, matching the structure of this paper. `summable_of_inputs` derives the conclusion from Lemmas 2.1 and 2.2, stated as hypotheses; `pairSieve` proves Lemma 2.2; and `selbergMeanSquare_of_RH` proves Lemma 2.1 from RH. The last of these is the main offensive, and its proof follows the classical route with two adaptations to the available infrastructure. The truncated explicit formula

$$\psi(y) = y - \sum_{|\gamma| \leq T} \frac{y^\rho}{\rho} + O\left(\frac{y \log^2(Ty)}{T} + \log^2(Ty)\right)$$

(cf. [2], Ch. 17) is obtained by moving the contour to $\operatorname{Re} s = -1/2$. On the left edge and the left halves of the horizontal edges, ζ'/ζ is controlled by reflecting through the functional equation, which needs a bound $|\psi_\Gamma(w)| \ll \log |\operatorname{Im} w|$ for the digamma function, proved from Euler's limit formula. The mean square of the zero sum is then computed as in [8], but against the polynomial weight $((u - X/2)(4X - u)/X^2)^2$ majorizing $1_{[X, 3X]}$. Two integrations by parts turn the kernel $\int u^{-1+i(\gamma-\gamma')} du$ into $O(1/(1 + (\gamma - \gamma')^2))$, which with the zero count $N(t + 1) - N(t) \ll \log t$ gives $hX \log^2 X$ without the extra logarithm a sharp cutoff would cost. The sieve generalizes the twin-type bound of the Carmichael formalization [1] to arbitrary intervals and linear forms. That formalization also supplies the Selberg sieve, the Perron kernel, contour integrals over rectangles, and zero counting, all reused verbatim. The new part is about 6,800 lines, and the reused part about 14,900. A kernel-checked test confirms that `PrimePrefixFree` selects exactly the 61 terms of A287117 below 536. The repository is <https://github.com/jdb19937/prime-prefix-free>.

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